
LAB 2 - Enterprise Real-Time Streaming Pipeline **using Eventstream**

1. Project Overview

Project Title

Real-Time Retail Sales Analytics using Microsoft Fabric Eventstream

2. Real-Time Corporate Scenario

A multinational retail company receives thousands of live sales transactions every minute from:

- POS systems
- mobile applications
- e-commerce websites
- billing systems

The business wants:

- live dashboards
- real-time monitoring
- instant analytics
- streaming ingestion
- operational visibility

The organization decides to implement:

Fabric Eventstream Architecture

to process streaming sales events in real time.

3. Corporate Architecture

Retail Applications



Real-Time Sales Events



Fabric Eventstream



Lakehouse (Streaming Layer)



Power BI Real-Time Dashboard

4. Learning Objectives

By the end of this lab you will learn:

- Create Eventstream
 - Configure real-time source
 - Stream live events
 - Configure Lakehouse destination
 - Store streaming events
 - Monitor streaming pipeline
 - Build real-time analytics architecture
 - Understand enterprise streaming concepts
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5. Enterprise Use Cases

This architecture is used in:

Industry	Use Case
Banking	Fraud detection
Retail	Live sales tracking
Telecom	Network monitoring
Healthcare	Patient telemetry
Logistics	Vehicle tracking

Manufacturing	IoT sensor analytics
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6. Prerequisites

Component	Status
Fabric Workspace	✓
Fabric Trial Enabled	✓
Lakehouse Created	✓

7. Standards Naming

Component	Naming Convention
Eventstream	es-retail-sales-dev
Lakehouse	lh-salesanalytics-dev
Table	bronze_realtime_sales

8. PHASE 1 — Create Real-Time Eventstream

Step 1 — Open Fabric Workspace

Navigate:

Workspace →Your Workspace

Expected Output:

Workspace dashboard opens successfully.

Step 2 — Create Eventstream

Click:

New Item

Search:

Eventstream

Select:

Eventstream

Step 3 — Configure Eventstream

Provide:

Field	Value
Name	es-retail-sales-dev

Click:

Create

Expected Output:

Eventstream canvas opens successfully.

9. PHASE 2 — Configure Real-Time Source

Step 4 — Add Source

Inside Eventstream canvas:

Click:

Add Source

Expected Output:

Source selection panel opens.

Step 5 — Select Sample Data Source

Choose: Sample Data

Step 6 — Configure Sample Data

Provide:

Field	Value
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Source Name	src-sales-events
Event Type	Stock Market

Click:

Add

Expected Output:

Live event stream starts generating.

PHASE 2B — Configure Real-Time Weather Streaming

Step 6A — Add Additional Real-Time Source

Inside Eventstream canvas:

Click:

Add Source

Expected Output:

Source configuration panel opens.

Step 6B — Select Real-Time Weather

Choose:

Real-Time Weather

Expected Output:

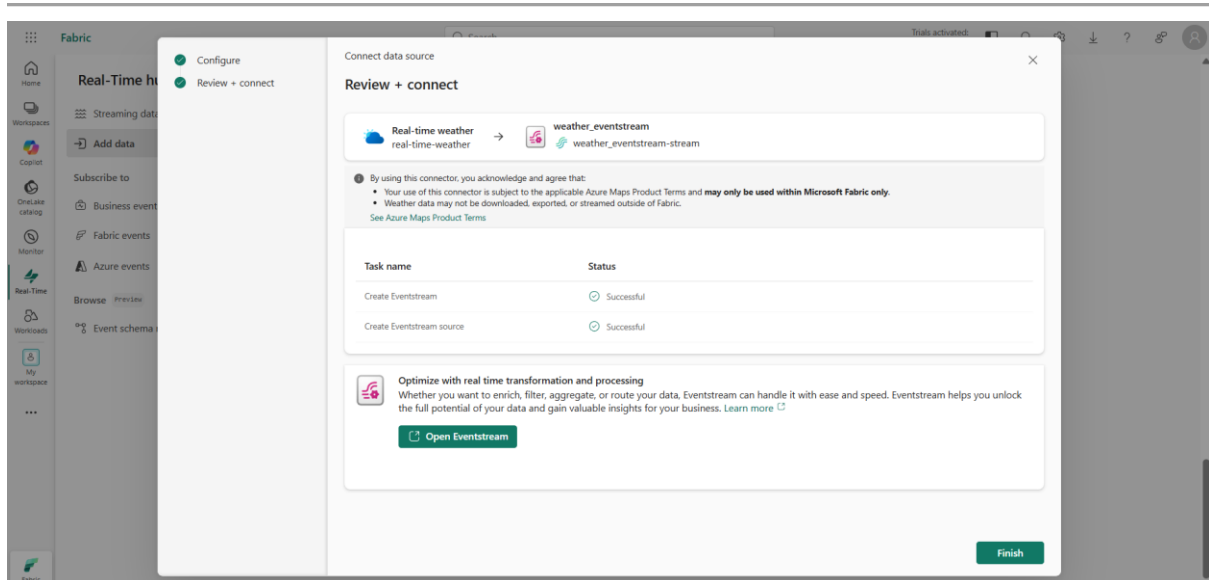
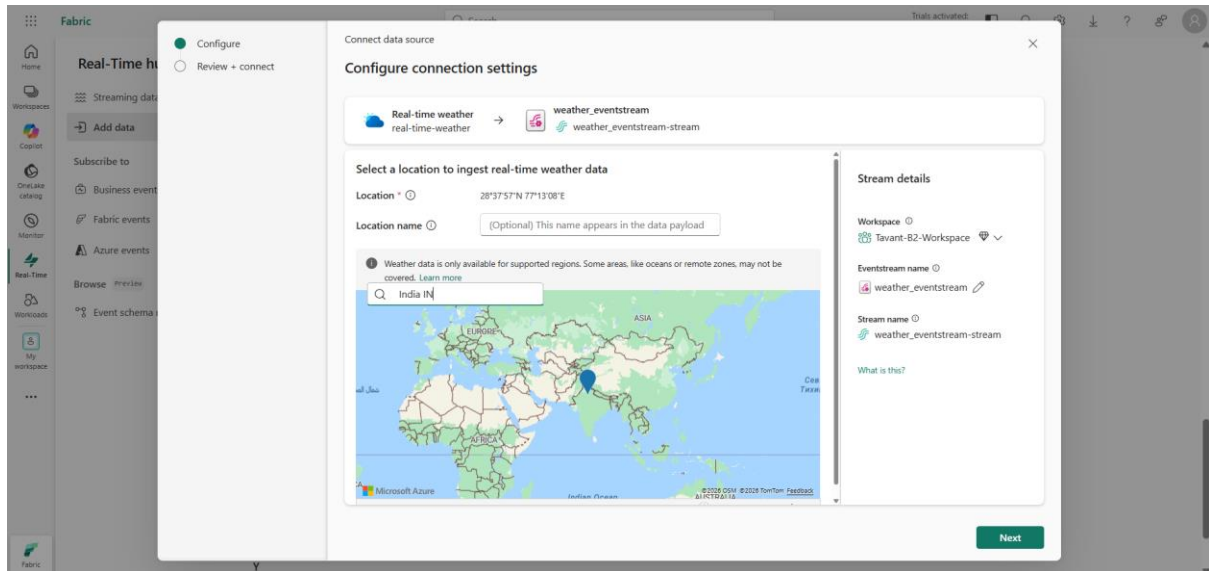
Weather streaming configuration page opens.

Step 6C — Configure Weather Source

Provide:

Field	Value
Location	INDIA

Click:
Next



Step 6D — Connect Weather Stream to Eventstream

Provide:

Field	Value
Eventstream	es-retail-sales-dev
Stream Name	weather-stream

Click:

Connect

Expected Output:

Real-time weather source connected successfully.

Step 6E — Validate Streaming Weather Events

Observe:

- temperature updates
- humidity changes
- wind speed
- live telemetry flow

Expected Output:

Continuous weather events visible in Eventstream.

The screenshot shows the Microsoft Fabric interface. On the left, the 'Data' pane is expanded, showing 'Sources' with 'real-time-weather' and 'Streams' with 'weather_eventstream'. The main canvas displays a data pipeline with three components: 'real-time-weather' (Source), 'weather_eventstream' (Stream), and 'Lakehouse' (Destination). All components are marked as 'Active'. Below the pipeline, the 'Data preview' tab is selected, showing a table of weather data.

dateTime	des...	ic...	h...	temperature	realFeelTemperature
2025-05-19T13:38:00+05:30	Hazy sunshine	("value":5)	false	("value":43.9,"unitLabel":"C","unitType":"value":17)	("value":45.2,"unitLabel":"C","unitType":"value":17)
2025-05-19T13:38:00+05:30	Hazy sunshine	("value":5)	false	("value":43.9,"unitLabel":"C","unitType":"value":17)	("value":45.2,"unitLabel":"C","unitType":"value":17)
2025-05-19T13:32:00+05:30	Hazy sunshine	("value":5)	false	("value":43.9,"unitLabel":"C","unitType":"value":17)	("value":45.2,"unitLabel":"C","unitType":"value":17)

10. PHASE 3 — Add Eventstream Destination

Step 7 — Add Destination

Click:

Add Destination

Expected Output:

Destination panel opens.

Step 8 — Select Lakehouse

Choose: Lakehouse

Step 9 — Configure Lakehouse Destination

Provide:

Field	Value
Destination Name	dest-sales-lakehouse
Workspace	Current Workspace
Lakehouse	lh-salesanalytics-dev
Table Name	bronze_realtime_sales

Data Mode: New Table

Click: Save

The screenshot displays the Microsoft Fabric interface for configuring a data pipeline. The pipeline is named 'es-retail-sales-dev' and is currently in 'Edit mode'. It consists of three components: 'StockMarket', 'es-retail-sales-dev', and 'Lakehouse'. The 'Lakehouse' component is highlighted with a red box and a warning icon, indicating it is not yet configured. The right-hand pane shows the configuration for the 'Lakehouse' component, with fields for 'Destination name' (dest-sales-lakehouse), 'Workspace' (Tavant-B2-Workspace), 'Lakehouse' (lh-salesanalytics-dev), 'Delta table' (dbo.bronze_realtime_sales), and 'Input data format' (Json). The 'Save' button is visible at the bottom of the configuration pane.

Expected Output:

Destination linked successfully.

11. PHASE 4 — Start Real-Time Streaming

Step 10 — Publish Eventstream

Click:

Publish

Expected Output:

Eventstream published successfully.

The screenshot displays the Microsoft Fabric workspace interface. On the left sidebar, the 'Data' section is expanded, showing 'Sources' (StockMarket), 'Streams' (es-retail-sales-dev), 'Operators' (No Operators), and 'Destinations' (dest-sales-lakehouse). The main canvas shows a data pipeline with three components: 'StockMarket' (Source), 'es-retail-sales-dev' (Eventstream), and 'dest-sales-lakehouse' (Destination). All components are marked as 'Active'. Below the canvas, the 'Data preview' tab is selected, showing a table of retail sales data. The table has columns for time, sector, stock type, and various metrics.

time	sector	stock type	bl...	bl...	as...	as...	la...	la...	v...	m...	
2026-05-19T08:11:02.765Z	HOOJ	mediaentertainment	commonstock	1330.28	22	0	0	1330.28	35	90989	0.0442
2026-05-19T08:11:02.765Z	BMZM	retailing	commonstock	2266.84	108	0	0	2246.84	27	202260	0.01773
2026-05-19T08:11:02.765Z	NSFT	softwareservices	commonstock	360.63	59	0	0	380.63	48	978670	0.02508
2026-05-19T08:11:02.765Z	HOOJ	mediaentertainment	commonstock	1330.28	106	0	0	1260.28	75	90961	0.0442
2026-05-19T08:11:02.764Z	BMZM	retailing	commonstock	2316.84	27	0	0	2356.84	57	185829	0.01773
2026-05-19T08:11:02.764Z	NSFT	softwareservices	commonstock	370.63	34	0	0	360.63	35	978623	0.02508
2026-05-19T08:11:02.764Z	HOOJ	mediaentertainment	commonstock	1280.28	27	0	0	1310.28	58	90948	0.0442
2026-05-19T08:11:02.764Z	BMZM	retailing	commonstock	2366.84	49	0	0	2276.84	12	174459	0.01773

Step 11 — Start Streaming

Eventstream automatically starts processing live events.

Expected Output:

Live events become visible on canvas.

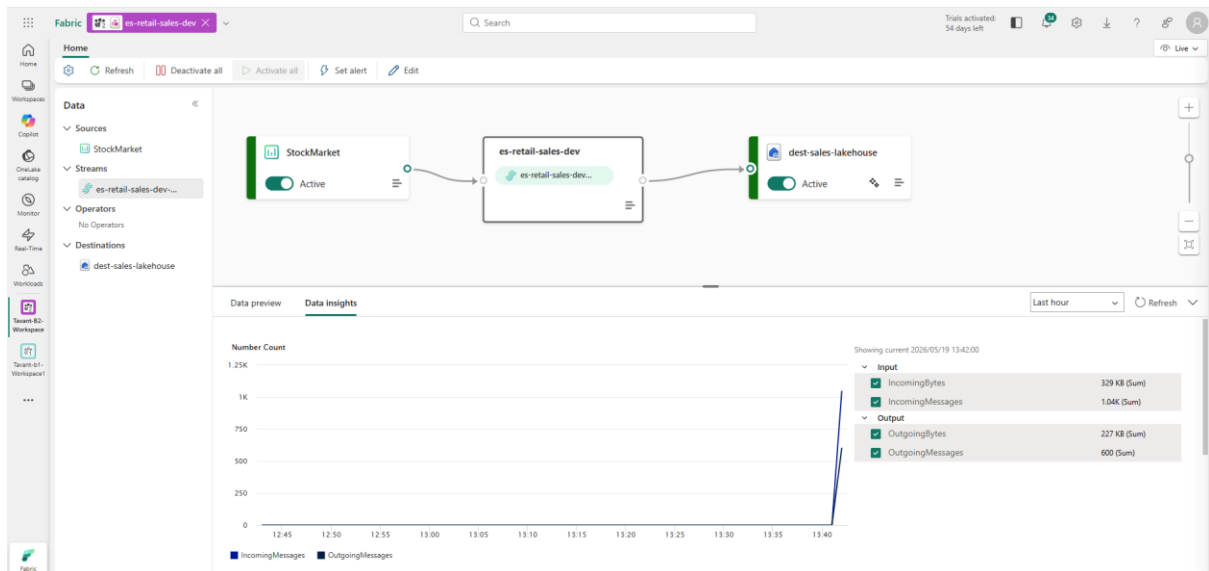
Step 12 — Monitor Event Flow

Observe:

- Incoming events
- Throughput
- Processing metrics

Expected Output:

Continuous event processing visible.



12. PHASE 5 — Validate Streaming Data in Lakehouse

Step 13 — Open Lakehouse

Navigate:

lh-salesanalytics-dev

Step 14 — Open Tables

Expand: Tables

Locate: bronze_realtime_sales

Expected Output: Streaming table created automatically.

	AM time	AM symbol	AM sector	AM security type	12 bidPrice	1% bidSize	1% askPrice	1% askSize	12 lastSalePrice	1% lastSaleSize	1% volume	12 marketPer...	1% seq
1	2026-05-19T08...	BMZM	retailing	commonstock	2386.84	80	0	0	2316.84	87	162086	0.01773	3029
2	2026-05-19T08...	HQOJ	mediaentertain...	commonstock	1350.28	86	0	0	1250.28	45	90975	0.0442	7947
3	2026-05-19T08...	NSFT	software services	commonstock	280.63	95	0	0	280.63	71	978621	0.02508	3059
4	2026-05-19T08...	BMZM	retailing	commonstock	2326.84	56	0	0	2256.84	91	182335	0.01773	6965
5	2026-05-19T08...	HQOJ	mediaentertain...	commonstock	1310.28	49	0	0	1290.28	93	90816	0.0442	1648
6	2026-05-19T08...	NSFT	software services	commonstock	380.63	51	0	0	320.63	49	978726	0.02508	5135
7	2026-05-19T08...	BMZM	retailing	commonstock	2286.84	116	0	0	2266.84	8	193551	0.01773	1323
8	2026-05-19T08...	HQOJ	mediaentertain...	commonstock	1320.28	23	0	0	1310.28	15	90906	0.0442	2798
9	2026-05-19T08...	NSFT	software services	commonstock	390.63	44	0	0	340.63	81	978794	0.02508	8289
10	2026-05-19T08...	BMZM	retailing	commonstock	2266.84	106	0	0	2326.84	80	203292	0.01773	4576
11	2026-05-19T08...	HQOJ	mediaentertain...	commonstock	1270.28	14	0	0	1260.28	95	90954	0.0442	1022
12	2026-05-19T08...	NSFT	software services	commonstock	390.63	122	0	0	410.63	59	978563	0.02508	9993
13	2026-05-19T08...	BMZM	retailing	commonstock	2276.84	57	0	0	2306.84	34	206426	0.01773	2649
14	2026-05-19T08...	HQOJ	mediaentertain...	commonstock	1270.28	110	0	0	1300.28	86	90861	0.0442	1486
15	2026-05-19T08...	NSFT	software services	commonstock	380.63	44	0	0	370.63	33	978680	0.02508	3498
16	2026-05-19T08...	BMZM	retailing	commonstock	2246.84	8	0	0	2326.84	98	180099	0.01773	6381
17	2026-05-19T08...	HQOJ	mediaentertain...	commonstock	1370.28	70	0	0	1250.28	67	90947	0.0442	1523
18	2026-05-19T08...	NSFT	software services	commonstock	290.63	16	0	0	310.63	22	978777	0.02508	4054
19	2026-05-19T08...	BMZM	retailing	commonstock	2346.84	41	0	0	2296.84	8	190371	0.01773	9090

13. PHASE 6 — Real-Time SQL Analytics

Step 15 — Open SQL Analytics Endpoint

Navigate:

SQL Analytics Endpoint

Step 17 — Run Streaming Query

Execute:

```
SELECT TOP 100 *  
FROM bronze_realtime_sales  
ORDER BY EventProcessedUtcTime DESC;
```

Expected Output:

Latest streaming events become visible.

Step 18 — Run Business Analytics Query

```
SELECT Product,  
COUNT(*) AS TOTAL_ORDERS  
FROM bronze_realtime_sales  
GROUP BY Product  
ORDER BY TOTAL_ORDERS DESC;
```

Expected Output:

Product	TOTAL_ORDERS
Laptop	250
Mobile	180

14. PHASE 7 — Build Real-Time Dashboard

Step 19 — Create Power BI Report

Inside Lakehouse:

Click:

New Report

Expected Output:

Power BI report canvas opens.

Step 20 — Add Real-Time Visuals

Create:

- Live sales KPI
- Product sales chart
- Regional analytics
- Streaming table view

Expected Output:

Dashboard starts refreshing with streaming data.

15. Enterprise Streaming Concepts

Real-Time vs Batch Processing

Real-Time	Batch
Continuous ingestion	Scheduled ingestion
Low latency	High latency
Streaming analytics	Periodic analytics

Eventstream Role

Eventstream acts as:

Streaming Ingestion Engine

Responsibilities:

- capture events
 - process streams
 - route events
 - deliver to destinations
-

16. Enterprise Medallion Architecture

Bronze Layer

Raw streaming ingestion.

bronze_realtime_sales

Silver Layer

Cleaned streaming data.

Examples:

- deduplication
 - validation
 - transformations
-

Gold Layer

Business-ready KPIs.

Examples:

- hourly revenue
 - customer analytics
 - product performance
-

Monitoring

Monitor:

- event lag
 - throughput
 - dropped events
 - processing latency
-

Security

Use:

- RBAC

- Managed Identity
- Workspace roles
- Data governance

Scalability

Eventstream supports:

- large-scale streaming
- multi-source ingestion
- enterprise observability

Q1 — What is Eventstream in Fabric?

Eventstream is a real-time ingestion and event processing service inside Microsoft Fabric.

Q2 — Difference Between Pipeline and Eventstream?

Pipeline	Eventstream
Batch processing	Real-time streaming
Scheduled execution	Continuous ingestion
ETL focused	Event driven

Q3 — Why Use Lakehouse as Destination?

Lakehouse supports:

- scalable analytics
- SQL querying
- Power BI integration
- medallion architecture

Q4 — What Are Streaming Use Cases?

- fraud detection
- IoT monitoring
- live dashboards
- operational analytics

-
- telemetry processing
-

21. Final Enterprise Architecture

Applications / Devices



Eventstream



Lakehouse Bronze Layer



SQL Analytics Endpoint



Power BI Real-Time Dashboard
